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夺命毒雾：伦敦1952

The Killer Smog of 1952

温暖的炉火为何变成致命烟雾？

英文原文

On Friday 5 December 1952, cold, **stagnant** air **descended on** London. **Overhead**, warmer air pressed down like a **lid**, holding the **chill captive** beneath. Londoners were **accustomed to murky**, **dreary** winters. This time, though, the capital's smoke was trapped too, thickening into a **choking smog**, a mix of **smoke** and **fog**.

That smoke, from coal burned in **domestic fireplaces** and **power stations**, was **laden with** soot and **noxious** sulphur dioxide. In places, **visibility** fell to a few metres. **Ambulance** services were **suspended**, an opera at the Sadler's Wells theatre was abandoned at the **interval**, and prize **cattle** at the Smithfield **agricultural** show died or had to be **slaughtered**.

The smog **dissipated** on Tuesday 9 December. Officials at the time put the **excess** deaths at about 4,000, but a 1954 Ministry of Health report blamed later deaths on **influenza**. A study published in Environmental Health Perspectives in 2001 **reassessed** the evidence and **cast doubt on** that **explanation**, estimating about 12,000 excess deaths between December 1952 and February 1953.

For decades Londoners had **tolerated** the smoke as a **nuisance**. One **villain flickered** in their own grates, a source of **warmth** and **comfort** so **habitual** that few thought to **accuse** it. Such **complacency** may help to explain the government's **grudging response**. It **withstood** pressure for an **inquiry** until May 1953.

Reform proved **protracted**, slowed by official **inertia**. The Clean Air **Act** became law on 5 July 1956, three and a half years after the smog. It **empowered municipal** councils to create **smokeless** zones, **combating** smoke at its **source**. The **cosy hearth**, **beloved of** British families, had, all along, been among the chief **offenders**.

中文译文

1952年12月5日星期五，寒冷、凝滞的空气笼罩降临了伦敦。头顶上方，较暖的空气压了下来，像盖子一样，把寒气困在下面。伦敦人早已习惯昏暗的、阴沉的冬天。不过这一次，首都的煤烟也被困住了，越积越浓，成了令人窒息的烟雾，也就是烟与雾的混合物。

这些煤烟来自于烧的煤，在家庭壁炉和发电厂里，充满了煤烟和有毒的二氧化硫。有些地方，能见度降到只有几米。救护车服务被暂停，萨德勒之井剧院的一场歌剧在幕间休息时被中止，而史密斯菲尔德农业展上的获奖牛则死亡或不得不被宰杀。

烟雾消散于12月9日，星期二。当时的官员估计超额死亡约4,000人，但卫生部1954年的一份报告把后来的死亡归因于流感。2001年发表于《环境健康展望》的一项研究重新评估了证据，并对这一解释提出了质疑，估计在1952年12月至1953年2月期间约有12,000例超额死亡。

几十年来，伦敦人一直容忍着煤烟，只是把它当成一种让人讨厌的事情。一个元凶就在他们自家的炉栅里闪烁，一个温暖与舒适的来源，如此习以为常，以至于很少有人想到去指责它。这种自满，或许有助于解释政府勉强的回应。政府顶住了要求调查的压力，直到1953年的5月。

改革被证明是旷日持久的，并被官方的惰性拖慢了。《清洁空气法》于1956年7月5日成为法律，此时距那场烟雾已过去三年半。该法授权市镇议会设立无烟区，从源头上遏制煤烟。英国家庭深受喜爱的、温馨的炉膛，原来一直都在主要的肇事者之列。

词汇与表达

stagnant adj.
凝滞的
5 December 1952, cold, stagnant air

descended on v.
笼罩降临了
stagnant air descended on London.

Overhead adv.
头顶上方

lid n.
盖子

chill n.
寒气

captive adj.
困

accustomed to adj.
习惯
accustomed to murky, dreary winters.

murky adj.
昏暗的
accustomed to murky, dreary winters.

dreary adj.
阴沉的

smoke n.
烟

choking adj.
令人窒息的
thickening into a choking smog, a mix

smog n.
烟雾
thickening into a choking smog, a mix

fog n.
雾

domestic adj.
家庭

fireplaces n.
壁炉

power stations n.
发电厂

laden with adj.
充满了
was laden with soot and noxious

noxious adj.
有毒的
with soot and noxious sulphur dioxide.

visibility n.
能见度
In places, visibility fell to a few

Ambulance n.
救护车

suspended v.
暂停
Ambulance services were suspended,

interval n.
幕间休息

cattle n.
牛

agricultural adj.
农业

slaughtered v.
宰杀

dissipated v.
消散
The smog dissipated on Tuesday 9

excess adj.
超额
at the time put the excess deaths at

influenza n.
流感

词汇与表达（续）

reassessed v.
重新评估了
reassessed the evidence

cast doubt on phr.
对……提出了质疑
and cast doubt on that explanation,

explanation n.
解释

tolerated v.
容忍

nuisance n.
让人讨厌的事情
had tolerated the smoke as a nuisance.

villain n.
元凶

flickered v.
闪烁

warmth n.
温暖

comfort n.
舒适

habitual adj.
习以为常
of warmth and comfort so habitual

accuse v.
指责

complacency n.
自满
Such complacency may help to explain

grudging adj.
勉强的
the government's grudging response.

response n.
回应

withstood v.
顶住了

inquiry n.
调查
for an inquiry until May 1953.

protracted adj.
旷日持久的
Reform proved protracted, slowed by

inertia n.
惰性
slowed by official inertia.

Act n.
法

empowered v.
授权
It empowered municipal councils to

municipal adj.
市镇

smokeless adj.
无烟

combating v.
遏制

source n.
源头

cosy adj.
温馨的

hearth n.
炉膛

beloved of adj.
深受喜爱的

offenders n.
肇事者

文章简述

- 困住：1952年12月5至9日，伦敦上方暖空气像盖子压住冷空气，煤烟无法散去，形成致命烟雾。
- 源头：家庭壁炉和发电厂烧煤，释放烟尘及有毒二氧化硫；熟悉的温暖炉火也是污染元凶。
- 瘫痪：局部能见度仅几米，救护车停运、演出中止，农业展的牛也死亡或被迫宰杀。
- 复核：官方初估比常态多死亡约4000人，后续死亡曾被归于流感；2001年研究重估1952年12月至1953年2月，多死亡约1.2万人。
- 改革：政府拖延调查与改革，直到三年半后的1956年才制定《清洁空气法》，授权设无烟区，从源头控烟。

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稻济天下

Rice for the World

杂交稻的丰收，怎样由许多人共同促成？

英文原文

In July 1964, **inspecting** rice ear by ear in a **paddy** at Anjiang Agricultural School in Hunan province, the **instructor** Yuan Longping spotted a **peculiar** plant. It was male-**sterile**, **shedding** no **viable pollen**. Rice **pollinates itself**, so breeding **hybrid** seed **in bulk** depends on such plants. In February 1966 he **detailed** the **anomaly** in the **journal** Chinese Science Bulletin.

The **specimen launched** a six-year **hunt**: Yuan and his **colleagues** could not **breed** a sterile line whose **offspring reliably inherited** the **trait**. The **breakthrough** came on 23 November 1970, in a **marsh** near Nanhong Farm in Sanya, Hainan. Yuan's assistant, Li Bihu, and Feng Keshan, a farm **technician**, found wild rice whose pollen was **abortive**.

Then came Yuan's **decisive** move: rather than **withhold** the **strain**, he **distributed** it **to collaborators nationwide**. Hybrids need three **distinct** lines: a sterile line, one to **maintain** its **sterility**, and one to **restore fertility** in the hybrids. Yan Longan's team in Jiangxi bred sterile lines and Zhang Xiancheng in Guangxi first **identified** a restorer line. In October 1973, at a **national conference** in Suzhou, Yuan **declared** the three-line system complete.

By 1976 hybrids were **under cultivation** on about 140,000 **hectares**. The World Food Prize Foundation **estimates** that hybrid rice can **yield** up to a fifth more **grain** than **conventional varieties**. By the early 2000s it **covered** almost half of China's **rice fields**.

Yuan, who shared the 2004 **World Food Prize** with Sierra Leone's Monty Jones, died in Changsha on 22 May 2021, aged 91. Yet popular **myth portrays** him **as** a **solitary genius**. His eye began the search; his **generosity** let many hands **bring in the harvest**.

中文译文

1964年7月，一穗一穗地检查水稻时，在湖南省安江农校的一块稻田里，教员袁隆平发现了一株奇特的/奇怪的稻子。它是雄性不育株，散播不出能生长发育的花粉。水稻自花授粉，所以，要大批地培育杂交种子，就得靠这样的植株。1966年2月，他详细描述了这一反常现象，文章发表在期刊《科学通报》上。

这株标本开启了一场长达六年的搜寻：袁隆平和同事们始终培育不出一个不育系，其后代能稳定地遗传这一性状。突破出现在1970年11月23日，地点是一片沼泽，在海南三亚南红农场附近。袁隆平的助手李必湖和农场技术员冯克珊找到了一种野生稻，它的花粉败育。

随后，袁隆平迈出了决定性的一步：而非扣留这个品系在自己手里，他把它分给全国各地的协作者。杂交稻需要三个各不相同的系：不育系，保持其不育性的保持系，以及使杂交种恢复育性的恢复系。颜龙安的团队在江西育成了不育系，张先程在广西率先找出了恢复系。1973年10月，在苏州召开的全国性会议上，袁隆平宣布三系已经配套。

到1976年，杂交水稻已经处于大面积种植，占地面积大约140000公顷。据世界粮食奖基金会估计，杂交稻比常规品种最多可多产五分之一的稻谷。到21世纪初，杂交稻已覆盖近一半的中国稻田。

袁隆平曾与塞拉利昂的蒙蒂·琼斯共获2004年世界粮食奖，2021年5月22日在长沙逝世，享年91岁。然而，一种流传甚广的错误的观点把他描绘成一个单枪匹马的天才。他的慧眼开启了这场搜寻；他的慷慨，让许多双手一起迎来了丰收。

词汇与表达

inspecting v.
检查
inspecting rice ear by ear

paddy n.
稻田

instructor n.
教员

peculiar adj.
奇特的/奇怪的
a peculiar plant

sterile adj.
不育
It was male-sterile

shedding v.
散播

viable adj.
能生长发育的
no viable pollen

pollen n.
花粉
no viable pollen

pollinates itself v.
自花授粉

in bulk adv.
大批地
breeding hybrid seed in bulk

hybrid n.
杂交
breeding hybrid seed in bulk

detailed v.
详细描述了

anomaly n.
反常现象
detailed the anomaly

journal n.
期刊

specimen n.
标本
The specimen launched a six-year hunt

launched v.
开启了

hunt n.
搜寻

colleagues n.
同事们

breed v.
培育
breed a sterile line

offspring n.
后代

reliably adv.
稳定地

inherited v.
遗传
reliably inherited the trait

trait n.
性状
reliably inherited the trait

breakthrough n.
突破
The breakthrough came

marsh n.
沼泽

technician n.
技术员

abortive adj.
败育
whose pollen was abortive

词汇与表达（续）

decisive adj.
决定性的
Yuan's decisive move

withhold v.
扣留
rather than withhold the strain

strain n.
品系

distributed to v.
分给

collaborators n.
协作者
collaborators nationwide

nationwide adj.
全国各地的

distinct adj.
各不相同的
three distinct lines

maintain v.
保持

sterility n.
不育性

restore v.
恢复
restore fertility in the hybrids

fertility n.
育性
restore fertility in the hybrids

identified v.
找出了

national conference n.
全国性会议

declared v.
宣布

under cultivation n.
种植

hectares n.
公顷

estimates v.
估计

yield v.
产
yield up to a fifth more grain

grain n.
稻谷

conventional adj.
常规

varieties n.
品种

covered v.
覆盖

rice fields n.
稻田

World Food Prize n.
世界粮食奖

myth n.
错误的观点

portrays as v.
把……描绘成

solitary adj.
单枪匹马的

genius n.
天才

generosity n.
慷慨
his generosity

bring in the harvest n.
迎来了丰收

文章简述

- 难题：水稻通常自己授粉，要大批量杂交，得用不产有效花粉的母株，让它接受另一种稻的花粉。袁隆平1964年找到这种稻，却难让后代稳定保持这一特性。
- 协作：1970年，李必湖、冯克珊找到关键野生稻。袁隆平把材料分给全国团队，颜龙安团队育成不育系，张先程率先找到恢复系。
- 配套：1973年三类稻株配齐：不育系当母株；保持系给它授粉，繁殖出仍不产有效花粉的母株；恢复系再与母株杂交，得到能正常结实的下一代。
- 推广：1976年已开始大面积种植，21世纪初覆盖中国近半稻田。世界粮食奖基金会估计，产量最多可比常规稻高两成。
- 贡献：成果来自发现、材料共享与全国协作。袁隆平2004年共获世界粮食奖，2021年去世。

炸药大王的救心药

Dynamite for the Heart

同一种液体，怎样既能炸山又能救心？

英文原文

Around 1847 the **Italian chemist** Sobrero first made **nitroglycerin**, a **colourless**, **oily liquid** of **acute instability**. A **mere droplet** on the tongue, he reported, left his head **throbbing violently** for hours. On 3 September 1864 a **nitroglycerin shed** in Stockholm exploded. The **tragic** blast killed five, including Alfred Nobel's younger brother Emil.

Nobel **tamed** the liquid by **blending** it with kieselguhr, a **porous** earth, into a **paste** shaped into **rods**. Patented in 1867 as dynamite, it was **harnessed** for **mining**, **quarrying**, and **demolition**, **enriching** him **immensely**. His 1895 **testament** made the **fortune** an **endowment** for prizes, first awarded in 1901.

Nitroglycerin had **side-effects**. **Factory operatives** suffered **racking** headaches, and their **tolerance**, built over the week, **ebbed** at weekends. Mondays brought a **recurrence**, with **dizziness** and **palpitations**, **hence** the name Monday disease. In 1879 William Murrell reported in The Lancet on its **therapeutic** use against angina pectoris, attacks of chest pain.

On 25 October 1896 Nobel, who had angina, wrote to his **aide** that **by an irony of fate** doctors had **prescribed** him **nitroglycerin** to take **internally**. Its **alias**, Trinitrin, was chosen, he explained, to avoid **frightening pharmacists** or the public. He did not **comply**. **Partly paralysed** by a **cerebral haemorrhage**, he died at his San Remo **villa** on 10 December 1896.

In 1977 Murad **ascertained** that **nitroglycerin** releases **nitric oxide**, which **widens** blood vessels. Furchgott found an **unidentified** relaxing **factor** in **vascular linings** in 1980, and Ignarro recognised it as **nitric oxide** in 1986. The **trio** shared the 1998 **Nobel Prize** in Physiology or Medicine. Its **presentation** speech, on 10 December, the **anniversary** of Nobel's death, **quoted** his **ironic** letter.

中文译文

1847年前后，意大利化学家索布雷罗首次制得硝酸甘油，这是一种无色、油状的液体，具有极度的不稳定性。仅仅一小滴放在舌头上，他报告说，就让他头部剧烈地抽痛了好几个小时。1864年9月3日，一间硝酸甘油棚屋在斯德哥尔摩发生爆炸。这场悲惨的爆炸炸死了五个人，其中包括阿尔弗雷德·诺贝尔的弟弟埃米尔。

诺贝尔驯服了这种液体，方法是把它与硅藻土这种多孔的土混合，制成糊状物，再塑造成棒条状。1867年作为炸药获得专利后，它被控制并且用来采矿、采石和拆除工程，使他变得极其富有。他在1895年立下的遗嘱使这一大笔钱成为一笔捐赠基金，用来设立奖项，奖项首次颁发于1901年。

硝酸甘油有副作用。工厂工人饱受折磨人的头痛，他们的耐受性在一周中逐渐建立，到周末又减退。每逢星期一，头痛便复发，还伴有眩晕和心悸，因此得名星期一病。1879年，威廉·默雷尔在《柳叶刀》上报告了它对心绞痛的治疗用途，也就是阵发性的胸痛。

1896年10月25日，患有心绞痛的诺贝尔写信给他的助手，说命运真是讽刺，医生竟给他开具了硝酸甘油让他内服。这种药的别名Trinitrin是特意选的，他解释说，为的是避免吓到那些药剂师和公众。他没有遵从医嘱。身体部分瘫痪，是由脑部出血造成的，他于1896年12月10日在圣雷莫的别墅中去世。

1977年，穆拉德查明硝酸甘油会释放一氧化氮，它会扩张血管。弗奇戈特发现了一种未经鉴定的舒张因子，存在于血管的内壁里，这是1980年的事；伊格纳罗认定它就是一氧化氮，这是1986年的事。这个三人组分享了1998年诺贝尔生理学或医学奖。这个奖的颁奖致辞在12月10日，也就是诺贝尔的逝世纪念日，引述了他那封感叹命运讽刺的信。

词汇与表达

Italian adj.
意大利

chemist n.
化学家
the Italian chemist Sobrero

nitroglycerin n.
硝酸甘油

colourless adj.
无色

oily adj.
油状的

liquid n.
液体

acute adj.
极度的
of acute instability

instability n.
不稳定性
of acute instability

mere adj.
仅仅

droplet n.
小滴

throbbing v.
抽痛了

violently adv.
剧烈地

shed n.
棚屋

tragic adj.
悲惨的

tamed v.
驯服了
Nobel tamed the liquid

blending v.
混合
blending it with kieselguhr

porous adj.
多孔的
a porous earth

paste n.
糊状物

rods n.
棒条

harnessed v.
控制并且用来
harnessed for mining

mining n.
采矿

词汇与表达（续）

quarrying n.
采石

demolition n.
拆除工程
mining, quarrying, and demolition

enriching v.
使……变得……富有
enriching him immensely

immensely adv.
极其

testament n.
遗嘱
His 1895 testament

fortune n.
一大笔钱
the fortune an endowment

endowment n.
捐赠基金
an endowment for prizes

side-effects n.
副作用

Factory operatives n.
工厂工人

racking adj.
折磨人的

tolerance n.
耐受性
their tolerance

ebbed v.
减退

recurrence n.
复发
a recurrence, with dizziness

dizziness n.
眩晕

palpitations n.
心悸

hence adv.
因此

therapeutic adj.
治疗
its therapeutic use

aide n.
助手

by an irony of fate phr.
命运真是讽刺

prescribed v.
开具了
doctors had prescribed him

词汇与表达（续）

internally adv.
内服

alias n.
别名

frightening v.
吓到

pharmacists n.
药剂师

comply v.
遵从
He did not comply

Partly adv.
部分

paralysed adj.
瘫痪

cerebral haemorrhage adj.
脑部出血

villa n.
别墅

ascertained v.
查明
Murad ascertained

nitric oxide n.
一氧化氮

widens v.
扩张

unidentified adj.
未经鉴定的
an unidentified relaxing factor

factor n.
因子

vascular adj.
血管的
vascular linings

linings n.
内壁

trio n.
三人组

Nobel Prize n.
诺贝尔……奖

presentation n.
颁奖

anniversary n.
纪念日
the anniversary of Nobel's death

quoted v.
引述了

ironic adj.
讽刺的

文章简述

- 危险：约1847年，索布雷罗制得极不稳定的硝酸甘油；1864年，一次爆炸夺去五人生命，包括诺贝尔的弟弟。
- 致富：诺贝尔把这种液体与硅藻土混合，制成较易使用的炸药，1867年获专利，积累的财富后来用于设立诺贝尔奖。
- 药用：接触硝酸甘油的工人会头痛，但医生又发现它能治疗心绞痛，1879年已有相关报告。
- 反差：诺贝尔晚年也患心绞痛，医生竟把他用来造炸药的同一种物质开给他治病。他写信感叹命运讽刺，也没有遵医嘱；同年因脑出血去世。
- 解释：后来研究查明，硝酸甘油会释放一氧化氮，使血管扩张。三位相关研究者获1998年医学诺奖，颁奖致辞在诺贝尔忌日引用了那封信。

被拒绝了十二次的哈利·波特

The Book Twelve Publishers Turned Down

十二次拒绝后，一位八岁读者带来了什么转机？

英文原文

In 1990, on a **delayed** train from **Manchester** to London's King's Cross, Joanne Rowling, a **would-be novelist**, was **unexpectedly** struck by the **inspiration** for Harry Potter, an **orphaned** young **wizard**. **Grief** soon **overshadowed** that **spark**: her mother died on 30 December 1990, aged 45. Rowling then **relocated** to Porto to teach English.

Rowling married in Porto in October 1992, and a daughter's **arrival** in July 1993 **ushered in motherhood**. At the end of 1993 she **resettled** in **Edinburgh**, a **lone** mother **reliant on** state **welfare**. She **habitually** wrote in cafés, **patiently polishing** her **unfinished** novel.

A **literary** agent, Christopher Little, **took on** the manuscript and **pitched** it **repeatedly** to **prospective** buyers. It was reportedly **rejected** by twelve publishers **in succession**, a **relentless** run of **refusals** that offered the **newcomer** little **encouragement**.

At Bloomsbury, the **chairman** Nigel Newton gave the first **chapter** to his eight-year-old daughter, Alice, who **relished** it and **eagerly** asked for the next **instalment**. The firm paid **a sum of** 1,500 pounds **as an advance**. **Fearful** that boys might **disregard** a book by a woman, the publishers **opted for initials** to **mask** her **gender**. Having no middle name, Rowling took the K for her **pseudonym** from her grandmother Kathleen.

Harry Potter and the Philosopher's Stone was published on 26 June 1997, in a **cautious** first print run of 500 **hardbacks**, 300 **allocated** to libraries. The American publisher Scholastic paid $105,000 for the US rights, and the **renamed** Sorcerer's Stone appeared there on 1 September 1998. The **series** and its **spin-off volumes** have **collectively surpassed** 500 million copies sold, in 80 languages. Alice's **appetite** was **evidently** shared by millions.

中文译文

1990年，在一列晚点的列车上，这列车从曼彻斯特开往伦敦国王十字车站，乔安妮·罗琳，一位有志成为小说家的人，出乎意料地得到了哈利·波特的灵感，哈利是一个成为孤儿的少年巫师。悲痛很快就盖过了那点火花：她的母亲于1990年12月30日去世，终年45岁。罗琳随后移居到波尔图，教英语。

罗琳于1992年10月在波尔图结婚；而1993年7月，女儿的降生开启/迎来了她为人母的生活。1993年底，她重新安家在爱丁堡，成了一位独自带孩子的母亲，依靠国家福利生活。她习惯在咖啡馆里写作，耐心地打磨她那部尚未完成的小说。

文学经纪人克里斯托弗·利特尔接下了这部书稿，并多次把它推销给潜在的买家。书稿据报道接连遭到十二家出版社拒绝，这一连串毫不留情的回绝，没给这位新人多少鼓励。

在布卢姆斯伯里出版社，董事长奈杰尔·牛顿把第一章交给了他八岁的女儿爱丽丝，她十分享受读这部小说，还急切地要看下一部分。这家出版社支付了一笔1,500英镑的款项作为预付金。由于担心男孩们可能会忽视女性写的书，出版方选择用名字缩写来掩饰/隐瞒她的性别。由于没有中间名，罗琳把 K用作她笔名的一部分，它取自祖母凯瑟琳的名字。

《哈利·波特与魔法石》（Philosopher's Stone）于1997年6月26日出版，首印十分谨慎，仅为500册精装本，其中300册分配给了图书馆。美国的学乐出版社以105,000美元买下了美国版权，而更名后的《魔法石》于1998年9月1日在那里问世。这个系列及其衍生的图书合计已超过了5亿册的销量，翻译成80种语言。爱丽丝的这份胃口，显然在数百万人中引发了共鸣。

词汇与表达

delayed adj.
晚点的

Manchester n.
曼彻斯特

would-be adj.
有志成为

novelist n.
小说家

unexpectedly adv.
出乎意料地

inspiration n.
灵感
the inspiration for Harry Potter

orphaned adj.
成为孤儿的

wizard n.
巫师

Grief n.
悲痛
Grief soon overshadowed that spark

overshadowed v.
盖过了
overshadowed that spark

spark n.
火花

relocated v.
移居
relocated to Porto

arrival n.
降生

ushered in v.
开启/迎来了
ushered in motherhood

motherhood n.
为人母

resettled v.
重新安家

Edinburgh n.
爱丁堡

lone adj.
独自

reliant on adj.
依靠
reliant on state welfare

welfare n.
福利
state welfare

词汇与表达（续）

habitually adv.
习惯

patiently adv.
耐心地

polishing v.
打磨
polishing her unfinished novel

unfinished adj.
尚未完成的

literary adj.
文学
A literary agent

took on phr.
接下了

pitched v.
推销
pitched it repeatedly

repeatedly adv.
多次

prospective adj.
潜在的
prospective buyers

rejected v.
拒绝
rejected by twelve publishers

in succession n.
接连
twelve publishers in succession

relentless adj.
毫不留情的

refusals n.
回绝

newcomer n.
新人

encouragement n.
鼓励

chairman n.
董事长

chapter n.
章

relished v.
十分享受
who relished it

eagerly adv.
急切地

instalment n.
部分

词汇与表达（续）

a sum of n.
一笔……款项

as an advance n.
作为预付金

Fearful adj.
担心

disregard v.
忽视
disregard a book by a woman

opted for v.
选择
opted for initials

initials n.
名字缩写
initials to mask her gender

mask v.
掩饰/隐瞒

gender n.
性别

pseudonym n.
笔名
the K for her pseudonym

cautious adj.
谨慎

hardbacks n.
精装本
first print run of 500 hardbacks

allocated v.
分配
300 allocated to libraries

renamed v.
更名后的

series n.
系列

spin-off n.
衍生的

volumes n.
图书

collectively adv.
合计
have collectively surpassed

surpassed v.
超过了
surpassed 500 million copies sold

appetite n.
胃口

evidently adv.
显然

文章简述

- 起笔：1990年，罗琳在一列晚点火车上得到哈利·波特的灵感，开始构思孤儿巫师的故事。
- 坚持：经历丧母后，她在爱丁堡独自抚养女儿，依靠福利生活，仍常在咖啡馆打磨小说。
- 转机：书稿据报遭十二家出版社拒绝；布卢姆斯伯里董事长的八岁女儿爱丽丝，读首章后急着看后续。
- 问世：该社接纳书稿，因担心男孩忽略女作者的书，采用J.K.缩写署名；1997年首册《魔法石》在英国首印500册精装本。
- 回响：美国版于1998年问世；系列及衍生图书后来已有80种语言版本，合计销量超过五亿册。

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从空气中造面包

The Bread Made from Air

同一种化学工艺，为何既能养人也能助战？

英文原文

In July 1909, BASF **envoys** gathered round a **squat** machine on a **bench** in Fritz Haber's Karlsruhe laboratory. Behind its **gleaming**, **bolted valves**, nitrogen and hydrogen met at **crushing** pressure and **furnace** heat, and **transparent beads** of **ammonia** began to **drip slowly**. Haber had **pried nourishment from** the air.

BASF's Carl Bosch **scaled** the method **up**, and from 1913 a **gigantic** plant at Oppau **mass-produced** ammonia. **Hailed for** making bread from air, Haber won the **coveted** 1918 Nobel Prize. **Ploughed into stony soils** as **fertiliser**, ammonia **wards off famine** as no **manure heap** could, bringing **abundance**. By 2008 it was feeding nearly half of **mankind**.

Haber's **ambition** had a **sinister** side. Ammonia **stocked** Germany's **arsenal with** nitrates for **munitions** despite a British **blockade**, and Captain Haber **marshalled** chemical warfare. On 22 April 1915 at Ypres, troops **vented** about 168 tonnes of **greenish**, **poisonous** chlorine towards the Allied **trenches**, the first large-scale gas **onslaught**.

Haber's wife, Clara Immerwahr, Germany's first woman with a **doctoral** degree in chemistry, voiced **objections to** his work. On 2 May 1915, after **festivities** in his honour, she shot herself in the garden with his service **pistol**. The next day the **widower** left for the Eastern Front.

Prestige offered no **sanctuary**. Of Jewish origin, Haber left Germany an **outcast** after Hitler's **regime seized** power, dying in Basel in 1934. His institute's **pesticide** research had produced Zyklon. A later **variant**, Zyklon B, was used in Nazi death camps, where relatives of his died.

About half the **nitrogen** in our **proteins** and DNA has **passed through** a Haber-Bosch factory, so each of us carries a little of Haber's chemistry.

中文译文

1909年7月，BASF的代表们聚集在一台矮胖的机器周围，机器就放在弗里茨·哈伯位于卡尔斯鲁厄的实验室里的一张工作台上。在它闪闪发亮、用螺栓紧固的阀门后面，氮和氢在压碎般的压力和熔炉般的高温下相遇，透明的氨（凝成的）液体小滴开始缓缓地滴落。哈伯从空气中撬出了养分。

BASF的卡尔·博施扩大了该方法的规模，自1913年起，奥保的一座庞大的工厂大批量生产氨。因“从空气中造面包”而广受赞誉，哈伯赢得了人人梦寐以求的令人垂涎的1918年诺贝尔奖。被翻耕进多石的土壤，作为化肥，氨抵御着饥荒，这是任何粪肥垒成的肥堆都做不到的，并带来丰饶。到2008年，它养活了将近一半的人类。

哈伯的雄心也有险恶的/邪恶的一面。氨用（制造）弹药所需的硝酸盐充实了德国的军火库，尽管有英国封锁，而哈伯上尉则组织调度了化学战。1915年4月22日，在伊普尔，（德国）部队朝协约国战壕方向排放了约168吨黄绿色、有毒的氯气——首次大规模毒气猛攻。

哈伯的妻子克拉拉·伊梅瓦尔，是德国第一位获得化学博士学位的女性，她曾对他的工作表达了反对。1915年5月2日，在为他致敬的庆祝活动之后，她在花园里用他的军用手枪自尽了。第二天，这位丧妻之人动身前往东线。

声望没有提供（给他）庇护。身为犹太裔，哈伯在希特勒政权夺取了权力之后，作为一个被驱逐的人离开了德国，并于1934年死于巴塞尔。他研究所的杀虫剂研究曾催生了齐克隆。后来的变体齐克隆B被用于纳粹死亡营，他的一些亲属就死在那里。

我们蛋白质和DNA中约有一半的氮，都经过了哈伯-博施工厂；因此，我们每个人体内都携带着一点哈伯的化学。

词汇与表达

envoys n.
代表

squat adj.
矮胖的

bench n.
工作台

gleaming adj.
闪闪发亮

bolted adj.
用螺栓紧固的

valves n.
阀门
its gleaming, bolted valves

crushing adj.
压碎般的

furnace n.
熔炉

transparent adj.
透明的

beads n.
液体小滴

ammonia n.
氨
transparent beads of ammonia

drip v.
滴落

slowly adv.
缓缓地

pried from v.
从……撬出了

nourishment n.
养分
pried nourishment from the air

scaled up 表达
扩大了……规模

gigantic adj.
庞大的

mass-produced v.
大批量生产
mass-produced ammonia

Hailed for 动词短语
因……广受赞誉

coveted adj.
人人梦寐以求的……令人垂涎的
the coveted 1918 Nobel Prize

Ploughed into v.
翻耕进

stony adj.
多石的

soils n.
土壤

fertiliser n.
化肥
as fertiliser

wards off v.
抵御
ammonia wards off famine

famine n.
饥荒
ammonia wards off famine

manure n.
粪肥

heap n.
肥堆

abundance n.
丰饶
bringing abundance

词汇与表达（续）

mankind n.
人类

ambition n.
雄心

sinister adj.
险恶的/邪恶的
a sinister side

stocked with v.
用……充实了

arsenal n.
军火库
Germany's arsenal

munitions n.
弹药
nitrates for munitions

blockade n.
封锁
despite a British blockade

marshalled v.
组织调度了
marshalled chemical warfare

vented v.
排放了

greenish adj.
黄绿色

poisonous adj.
有毒的

trenches n.
战壕

onslaught n.
猛攻
the first large-scale gas onslaught

doctoral adj.
博士

objections to n.
对……反对
voiced objections to his work

festivities n.
庆祝活动

pistol n.
手枪

widower n.
丧妻之人

Prestige n.
声望

sanctuary n.
庇护
Prestige offered no sanctuary

outcast n.
被驱逐的人
left Germany an outcast

regime n.
政权

seized v.
夺取了

pesticide n.
杀虫剂
His institute's pesticide research

variant n.
变体
A later variant, Zyklon B

nitrogen n.
氮
nitrogen and hydrogen met

proteins n.
蛋白质

passed through 动词短语
经过了

文章简述

- 突破：1909年，哈伯把空气中的氮与氢在高温高压下合成氨；博施将这套方法扩大到工厂，1913年实现大量生产。
- 养育：氨制成化肥，帮助粮食增产；到2008年，靠这种方法生产的化肥已养活近半人类，人体蛋白质和DNA中约一半的氮也经此工艺而来。
- 战争：合成氨也为德国弹药提供原料；哈伯还在1915年组织首次大规模毒气攻击，他的妻子反对这类战争工作。
- 命运：妻子后来开枪自杀，哈伯次日仍赴前线；他虽获1918年诺奖，却因犹太血统遭纳粹排斥，1934年死于流亡中。
- 余波：哈伯研究所研制的杀虫剂齐克隆，后来衍生出齐克隆B，被纳粹用于死亡营；他的一些亲属也死在这些营中。

失败的胶水，成功的便利贴

The Glue That Failed Its Way to Success

黏得不牢的胶，为什么成了畅销文具？

英文原文

In 1968 Spencer Silver, a scientist at 3M, **set out to** make a **super-strong adhesive**. The **polymer** he produced did **something else**. Its tiny **spheres** formed a layer with little **tack**, giving paper a **temporary** hold, so it would **adhere**, **peel off**, and be **reapplied**. To an **inventor** chasing strength, the **formulation** was a **setback**.

For five years Silver **acted as** its **promoter** within 3M's **corporate** ranks, in **informal** chats and at **seminars**. In his own **summary**, it was a **solution** without a problem. His **perseverance** met **indifference**. The **unconventional** material **found no use**, and the **pursuit** proved **fruitless**.

In 1974 Art Fry, a colleague who had heard one of those talks, faced a small **annoyance** at church. The paper **bookmarks wedged** in his **choir**'s **hymn** book kept getting **dislodged** and **falling out**. The **frustration** gave Fry an idea. The **unwanted** adhesive could make a marker that would **cling to** a page, **stay put**, and leave it **undamaged**.

In 1977 the **notes** were **test-marketed** as Press 'n Peel in four American cities, where **purchasers** showed little **enthusiasm**. In 1978 3M's **marketers staged** a **blitz** in Boise, Idaho, and **dispensed complimentary packets** of notes to offices. More than 90 per cent of **respondents** who tried them said they would buy. After the **coolness** came a **turnaround**.

On 6 April 1980 the **pads**, **rebranded as Post-it Notes**, went on sale **nationally** in the US, and Canada and Europe followed in 1981. The **stationery staple** owes its **signature canary tone** to an **unplanned** twist. The laboratory next door had only **leftover** yellow paper.

中文译文

1968年，斯潘塞·西尔弗，3M公司的一位科学家，着手研制一种超强黏合剂。然而他制成的聚合物却做了其他的事情。它的微小的球状颗粒形成一层黏着力很小的涂层，让纸张获得暂时的黏着，因此纸张能粘附上去，揭下后还能重新贴上。对一个追求强度的发明者来说，这种配方是一次挫折。

五年间，西尔弗在3M的企业上下充当它的推广者，在非正式的闲谈中、也在研讨会上介绍它。用他自己的话总结，这完全是一个解决“不存在的问题”的方案。他的坚持不懈换来的是冷漠。这种非常规的材料找不到用途，这番追求终究一无所获。

1974年，阿特·弗莱，一位听过其中一次讲座的同事，在教堂里遇到了一件小小的烦心事。他所在唱诗班的赞美诗集里夹着的纸书签总是移位，掉出来。这份懊恼让弗莱想到一个主意。这种没人要的黏合剂可以做成一种书签，附着在书页上不会移动，还能让书页完好无损。

1977年，这种便条以 Press 'n Peel 之名在美国四个城市试销，然而在这四个城市，购买者展示出极少的热情。1978年，3M的营销人员在爱达荷州博伊西发起一场闪电攻势，向很多办公室分发免费的一小包一小包的便条。试用过的受访者中，超过90%表示会购买。冷遇之后，迎来了转机。

1980年4月6日，这种便笺本更名为便利贴，在美国全国上市；1981年，便利贴又进入加拿大和欧洲。这款文具的必备品那标志性的金丝雀黄色调，源于一次计划外的转折。因为当时隔壁的实验室里，只有一些剩余的黄纸。

词汇与表达

set out to v.
着手

super-strong adj.
超强

adhesive n.
黏合剂
a super-strong adhesive

polymer n.
聚合物
The polymer he produced

something else pron.
其他的事情

spheres n.
球状颗粒

tack n.
黏着力
with little tack

temporary adj.
暂时的
a temporary hold

adhere v.
粘附
adhere, peel off, and be reapplied

peel off v.
揭下
adhere, peel off, and be reapplied

reapplied v.
重新贴上

inventor n.
发明者

formulation n.
配方
the formulation was a setback

setback n.
挫折
the formulation was a setback

acted as v.
充当

promoter n.
推广者

corporate adj.
企业

informal adj.
非正式的

seminars n.
研讨会

summary n.
总结

solution n.
解决……方案

词汇与表达（续）

perseverance n.
坚持不懈
His perseverance met indifference

indifference n.
冷漠
His perseverance met indifference

unconventional adj.
非常规的
The unconventional material

found no use v.
找不到用途

pursuit n.
追求

fruitless adj.
一无所获
the pursuit proved fruitless

annoyance n.
烦心事

bookmarks n.
书签

wedged v.
夹着的

choir n.
唱诗班

hymn n.
赞美诗

dislodged v.
移位
getting dislodged and falling out

falling out v.
掉出来

frustration n.
懊恼
The frustration gave Fry an idea

unwanted adj.
没人要的

cling to v.
附着在
cling to a page

stay put v.
不会移动

undamaged adj.
完好无损

notes n.
便条

test-marketed v.
试销
the notes were test-marketed

purchasers n.
购买者

词汇与表达（续）

enthusiasm n.
热情
purchasers showed little enthusiasm

marketers n.
营销人员

staged v.
发起

blitz n.
闪电攻势

dispensed v.
分发
dispensed complimentary packets

complimentary adj.
免费的
complimentary packets of notes

packets n.
小包

respondents n.
受访者
respondents who tried them

coolness n.
冷遇

turnaround n.
转机
After the coolness came a turnaround

pads n.
便笺本

rebranded as v.
更名为

Post-it Notes n.
便利贴

nationally adv.
全国

stationery n.
文具的

staple n.
必备品
The stationery staple

signature adj.
标志性的

canary n.
金丝雀黄

tone n.
色调

unplanned adj.
计划外的

leftover adj.
剩余的

文章简述

- 意外：1968年，3M的西尔弗想造超强胶，却做出由微小球粒组成的弱黏胶，纸张能贴住、揭下再贴。
- 用途：他在公司推广多年后，同事弗莱于1974年想到，用这种胶固定总从赞美诗集中滑落的书签，且不伤书页。
- 冷遇：1977年，这种便条在美国四个城市试销，消费者反应冷淡。
- 转机：1978年，3M在博伊西向办公室免费发试用品；试用过的受访者中，超过九成表示愿意购买。
- 上市：1980年，产品改名“便利贴”在全美上市，次年进入加拿大和欧洲；标志性的黄色，仅因隔壁实验室剩下黄纸。

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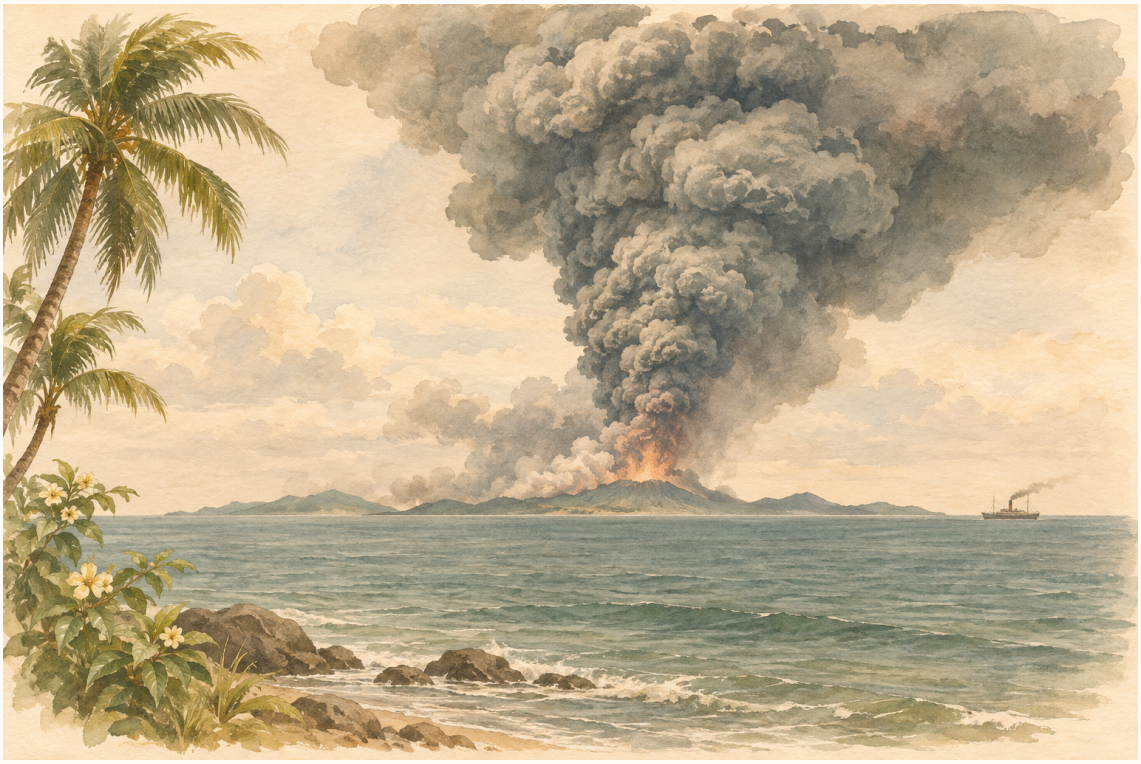
震动世界的一声巨响

The Sound That Shook the World



火山的一声巨响，如何传遍世界？

英文原文

Shortly after ten in the morning on 27 August 1883, Krakatoa, a **volcanic** island in the Sunda **Strait**, **unleashed** its most **thunderous** explosion. About four hours later, **islanders** on Rodrigues, nearly 4,800 kilometres across the Indian Ocean, took the **roar** for **distant gunfire**. **Barometers** worldwide recorded the **atmospheric** pressure wave, some of them seven times in about five days.

Restless since 20 May, the volcano reached a **catastrophic climax** on 26–27 August. About two-thirds of the island **collapsed into** its **emptied magma** chamber, **vanishing** beneath the sea. **Tsunamis**, **reportedly** over 30 metres high in places, **swamped coastal settlements** on Java and Sumatra. Dutch **colonial administrators** counted 36,417 dead, mostly **drowned**, perhaps **underestimating** the **toll**.

Telegrams from Batavia (now Jakarta), **relayed** by **submarine cable**, reached London within hours. Western newspapers soon began **chronicling** what is often called the first global news event. Yet early **dispatches** were **terse**; the **magnitude** of the **devastation** emerged **piecemeal** over days.

Ash and **sulphur** dioxide, **hurled** tens of kilometres high, formed a **haze** that circled the **equator** in about two weeks. In November 1883, **crimson** sunsets **glowed** over Europe and North America, **summoning firefighters** in New York and Connecticut to **phantom blazes**. Later climate **reconstructions** suggest global temperatures **dipped** a few tenths of a degree the following year.

英文原文（续）

Some experts **discounted** the volcanic explanation, blaming the **sunsets** on **moisture at high altitude** or **cosmic** dust. In February 1884, Britain's **Royal Society appealed** in The Times for dated observations, **eliciting** worldwide replies; the Society's 1888 report **tied** the glow **to** Krakatoa. Through barometers, cables, and crimson skies, **far-flung observers** began to **glimpse** how **interconnected** the planet already was.

中文译文

1883年8月27日上午10时刚过，喀拉喀托，一座位于巽他海峡的火山岛，迸发出其最为震耳欲聋的一声爆炸。约四小时后，罗德里格斯岛上的岛民，离着将近4800公里，远隔着一个印度洋，把这声轰鸣当成了远处的炮火声。世界各地的气压计都记录到了这次大气压力波，其中一些在大约五天之内记录到七次。

自5月20日起便躁动不安，这座火山达到灾难性的高潮是在8月26日至27日。约三分之二的岛体塌入已被排空的岩浆房，消失在海面之下。海啸据报道局部高逾30米，淹没了沿海居民点，遍及爪哇岛和苏门答腊岛。荷兰殖民地行政官员统计出36,417名死者，其中大多数是溺亡的，而他们或许低估了死亡人数。

电报从巴达维亚（今雅加达）发出，经海底电缆转发，在几小时内到达了伦敦。西方报纸很快便开始记述这起常被称为首个全球新闻事件的灾难。然而，早期的电讯都十分简短；浩劫的规模一点一点地显露出来，前后历经数日。

火山灰和二氧化硫被抛射到数十公里的高空，形成一层薄雾，在大约两个星期内环绕了赤道一周。1883年11月，深红色的晚霞发出暗淡的光，照耀着欧洲和北美，召唤着纽约州和康涅狄格州的消防员赶去扑救子虚乌有的大火。后来的气候重建研究推测，全球气温下降了零点几度，就在次年。

一些专家不予采信火山之说，而把晚霞归因于高空的水汽或宇宙尘埃。1884年2月，英国皇家学会发出呼吁，在《泰晤士报》上征集注明日期的观测记录，引起了世界各地的回应；学会1888年发表的报告把这片红霞归于喀拉喀托火山。借助气压计、电缆和深红色的天空，相隔遥远的观察者们开始认识到，这颗星球上的各个地方早已相互关联在一起。

词汇与表达

volcanic adj.
火山

Strait n.
海峡

unleashed v.
迸发出
unleashed its most thunderous

thunderous adj.
震耳欲聋的
its most thunderous explosion.

islanders n.
岛民

roar n.
轰鸣

distant adj.
远处的

gunfire n.
炮火声

Barometers n.
气压计
Barometers worldwide recorded the

atmospheric adj.
大气
the atmospheric pressure wave,

Restless adj.
躁动不安

catastrophic adj.
灾难性的
a catastrophic climax on 26–27 August.

climax n.
高潮
a catastrophic climax on 26–27 August.

collapsed into v.
塌入
the island collapsed into its emptied

emptied adj.
已被排空的

magma n.
岩浆

vanishing v.
消失

Tsunamis n.
海啸

reportedly adv.
据报道
Tsunamis, reportedly over 30 metres

swamped v.
淹没了
swamped coastal settlements on Java

词汇与表达（续）

coastal adj.
沿海

settlements n.
居民点

colonial adj.
殖民地

administrators n.
行政官员

drowned v.
溺亡

underestimating v.
低估了
perhaps underestimating the toll.

toll n.
死亡人数
perhaps underestimating the toll.

Telegrams n.
电报

relayed v.
转发
relayed by submarine cable,

submarine adj.
海底

cable n.
电缆

chronicling v.
记述

dispatches n.
电讯

terse adj.
简短
Yet early dispatches were terse;

magnitude n.
规模
the magnitude of the devastation

devastation n.
浩劫
the devastation emerged piecemeal over

piecemeal adv.
一点一点地
the devastation emerged piecemeal over

Ash n.
火山灰

sulphur n.
硫

hurled v.
抛射

haze n.
薄雾
haze that circled the equator in about

词汇与表达（续）

equator n.
赤道

crimson adj.
深红色的

glowed v.
发出暗淡的光

summoning v.
召唤着

firefighters n.
消防员

phantom adj.
子虚乌有的

blazes n.
大火

reconstructions n.
重建
Later climate reconstructions suggest

dipped v.
下降了

discounted v.
不予采信
Some experts discounted the volcanic

at high altitude n.
高空

sunsets n.
晚霞

moisture n.
水汽

cosmic adj.
宇宙

Royal Society n.
皇家学会

appealed v.
呼吁

eliciting v.
引起了
eliciting worldwide replies;

tied to v.
把……归于

far-flung adj.
相隔遥远的
far-flung observers began to glimpse

observers n.
观察者

glimpse v.
认识到

interconnected adj.
相互关联
how interconnected the planet already

文章简述

- 巨响：1883年喀拉喀托火山喷发，声传近4800公里；全球气压计录到压力波，有些约5天记录到它经过7次。
- 灾难：约三分之二岛体塌入地下原先储存岩浆、此时已排空的空间，并没入海中；海啸据报局部逾30米，官方记36417人死亡，多数溺亡，或仍低估。
- 快讯：海底电缆让电报数小时到伦敦，消息迅速传遍全球；但早期报道简短，灾情规模数日后才逐渐清楚。
- 余波：火山灰和二氧化硫形成高空薄雾，远方出现红霞；后人根据气候证据估算，1884年全球降温零点几度。
- 考证：当时有人把红霞归于高空水汽或宇宙尘埃；皇家学会汇集全球观测，1888年报告将红霞与喷发联系起来。

光之殇

Lost in the Glare

昆虫绕着灯飞，真只是因为喜欢光吗？

英文原文

A moth **skims over** an ultraviolet lamp and flips upside down, **belly** to the sky. In Costa Rica's **misty** Monteverde **cloud-forest**, **stereo** cameras caught such **acrobatics** for a 2024 Imperial College study that **unravelled** an **age-old** puzzle, the **helpless swirl** of insects round **porch** lights.

Folk wisdom blames the circling on **attraction**. In fact, insects **instinctively** keep their backs to whatever shines brightest, normally the sky, to stay level. A lamp, a **counterfeit** sky, **hijacks** this **built-in** habit, leaving insects **eddying** in its **halo** or **inverted** above it. In 477 field **clips**, **precious few darted** straight at the light.

Many fireflies **flirt** with their **lanterns**. A male **blinks** an **overture**, a female **winks** back, and bright nights **smother** the **banter**. At Tufts University, **suitors** under artificial light **mustered** about five flash patterns a minute instead of ten. Females **reciprocated** about half as often under red or blue light, scarcely at all under white, and never under **amber**.

The harm **spills into** the **ecosystem**. Along 55 Thames Valley **roadsides**, **lamp-lit hedges** held 47% fewer moth **caterpillars** than dark ones, with **harsh** white LEDs worse than **mellower sodium lamps**. Street lamps **rigged up** in seven of 14 Swiss **meadows** cut **nocturnal** pollinator visits by 62%. The plants set fewer fruit, and daytime pollinators proved poor **understudies**.

Over 80% of **humanity** lives beneath a **luminous veil**, which hides the pale **ribbon** of the Milky Way from more than a third. **Citizen** scientists' star counts from 2011 to 2022 show the sky brightening about 9.6% a year. At that **pace**, a child born where 250 stars **twinkle** would see about 100 by their eighteenth birthday.

中文译文

一只飞蛾掠过一盏紫外灯，随即上下颠倒地翻过身来，肚皮朝天。在哥斯达黎加蒙特韦尔德雾气缭绕的云雾林里，立体摄像机为伦敦帝国理工学院2024年的一项研究拍下了这类杂技动作。这项研究解开了一个由来已久的谜团，那就是昆虫为什么会身不由己地绕着门廊灯打转。

民间的老说法把这种打转归咎于吸引。事实上，昆虫会本能地让背部朝向最亮的东西，那通常是天空，以此保持平稳。灯成了一片冒牌的天空，劫持了这种与生俱来的习性，让昆虫在它的光晕里打着旋，或在它上方翻转过来。在477段野外视频片段中，径直冲向灯光的昆虫寥寥无几。

许多萤火虫用自己的灯笼来调情。雄虫闪烁着奏出一段序曲，雌虫眨眼般回应，而太亮的夜晚会淹没这番调笑。在塔夫茨大学，处在人造光下的求偶者每分钟只勉强打出约五组闪光，而不是十组。雌虫在红光或蓝光下回应的次数约减半，在白光下几乎不回应，在琥珀色光下则从不回应。

危害还会蔓延到整个生态系统。在泰晤士河谷的55处路边，有灯照明的树篱里，蛾类毛虫比没灯的地方少47%，而刺眼的白色LED比更柔和的钠灯更糟。在瑞士14片草甸中的7片，临时架起的路灯让夜间传粉者的造访减少了62%。植物结的果实变少了，而白天的传粉者也只是不称职的替补。

全球80%以上的人类生活在一层发光的帷幕之下，它让三分之一以上的人看不见银河那条淡淡的光带。公民科学家2011年至2022年的数星记录显示，夜空每年约变亮9.6%。按这个速度，若一个孩子出生在能看到250颗星星闪耀的地方，到18岁生日时只能看到约100颗。

词汇与表达

skims over v.
掠过
A moth skims over an ultraviolet lamp

belly n.
肚皮

misty adj.
雾气缭绕的

cloud-forest n.
云雾林

stereo adj.
立体

acrobatics n.
杂技

unravelled v.
解开了
unravelled an age-old puzzle

age-old adj.
由来已久的
an age-old puzzle

helpless adj.
身不由己地

swirl n.
打转

porch n.
门廊

Folk wisdom n.
民间的老说法
Folk wisdom blames the circling

attraction n.
吸引
blames the circling on attraction

instinctively adv.
本能地
insects instinctively keep their backs

counterfeit adj.
冒牌的
a counterfeit sky

hijacks v.
劫持了
hijacks this built-in habit

built-in adj.
与生俱来的
this built-in habit

eddying v.
打着旋

halo n.
光晕

inverted adj.
翻转
inverted above it

clips n.
视频片段

precious few phr.
寥寥无几
precious few darted straight

darted v.
径直冲

flirt v.
调情

lanterns n.
灯笼

blinks v.
闪烁

overture n.
序曲

词汇与表达（续）

winks v.
眨眼

smother v.
淹没

banter n.
调笑

suitors n.
求偶者

mustered v.
勉强打出

reciprocated v.
回应
Females reciprocated

amber n.
琥珀色

spills into v.
蔓延到

ecosystem n.
生态系统
The harm spills into the ecosystem

roadsides n.
路边

lamp-lit adj.
有灯照明的

hedges n.
树篱

caterpillars n.
毛虫
47% fewer moth caterpillars

harsh adj.
刺眼的
harsh white LEDs

mellower adj.
更柔和的

sodium lamps n.
钠灯
mellower sodium lamps

rigged up v.
临时架起
Street lamps rigged up

meadows n.
草甸

nocturnal adj.
夜间
nocturnal pollinator visits

understudies n.
替补
proved poor understudies

humanity n.
人类

luminous adj.
发光的
a luminous veil

veil n.
帷幕

ribbon n.
光带

Citizen n.
公民
Citizen scientists' star counts

pace n.
速度
At that pace

twinkle v.
闪耀

文章简述

- 飞行：昆虫本来靠把背部朝向明亮天空保持平衡。2024年研究发现，路灯成了“假天空”，昆虫把背转向灯，反而绕圈甚至翻身。
- 求偶：萤火虫靠闪光寻找配偶。实验中，人造光使雄虫发光减少，雌虫也更少回应，彼此传递的求偶信号受到干扰。
- 幼虫：英国路边调查发现，有灯树篱的蛾类毛虫比暗处少47%；白色LED的影响比钠灯更重。
- 传粉：瑞士实验中，有路灯的草甸比无灯草甸少了62%的夜间传粉昆虫来访，植物结出的果实也更少；白天的昆虫没能补上这份损失。
- 夜空：全球超过八成人口生活在光污染下，超过三分之一看不到银河。灯光不只遮住星星，也在改变昆虫的飞行、繁殖和植物的传粉。

冰海求生：坚忍号传奇

Survival in the Ice: The Legend of Endurance

冰海中的精确计算，怎样换来生路？

英文原文

On 5 December 1914 Endurance **departed** South Georgia, a **remote whaling outpost**, **for** Antarctica's Weddell Sea. In January 1915 ice **trapped** the **vessel**. On 27 October, her **hull buckling**, the men **abandoned** ship. She **foundered** on 21 November; her captain, Frank Worsley, **logged** the spot.

The **stranded** men **endured** over five **gruelling** months on the ice. Their **floe** split on 9 April 1916; on 15 April their **lifeboats** reached Elephant Island, a **bleak**, **uninhabited** rock far from **shipping lanes**.

On 24 April Shackleton and five others **embarked in** the James Caird, a **fragile** lifeboat, **on** a **perilous voyage** of some 1,300 kilometres. As **ferocious gales battered** the boat, Worsley **snatched** only four **sights of the sun**, **deducing** the rest by **dead reckoning** from **course**, speed, and **time elapsed**. On 10 May they **landed on** South Georgia's **deserted** south coast.

Shackleton, Worsley, and Tom Crean **traversed** its **mountainous interior** in about 36 **exhausting** hours, reaching the Stromness whaling station on 20 May. Ice **repelled** three **rescue attempts** before the Yelcho, a Chilean **tug**, **evacuated** the 22 **marooned** men on 30 August. All 28 of Endurance's party survived their **ordeal**.

On 5 March 2022 searchers **located** the **wreck** 3,008 metres down, **upright** and **intact**. She lay about four miles south of the spot Worsley recorded in November 1915, months before the voyage: a slight **discrepancy** for a **calculation** on **shifting** ice.

Legend **rightly salutes** Shackleton **for** his **charisma**. Yet both ends of the story **hinge on** Worsley's **meticulous arithmetic**. **Leadership held** 28 men **together**. **Precision** put six **ashore** within reach of help and, a century on, **brought** their ship **back** into view.

中文译文

1914年12月5日，坚忍号驶离了南乔治亚岛，一处偏远的捕鲸前哨，开往南极洲的威德尔海。1915年1月，冰困住了这艘船。10月27日，船体受压弯曲，众人弃船。它沉没于11月21日；它的船长弗兰克·沃斯利把沉没地点记入了航海日志。

被困住的众人熬过了五个多月艰苦累人的冰上日子。他们脚下的浮冰裂开了，那天是1916年4月9日；4月15日，他们的救生艇抵达了象岛，一座荒凉、无人居住的岩岛，远离航线。

4月24日，沙克尔顿和另外五人登上了詹姆斯·凯尔德号，一艘单薄脆弱的救生艇，踏上一段极度危险的航程，约1,300公里。当狂暴的/猛烈的大风多次猛击小艇的时候，沃斯利抓住了仅仅四次测日机会，推断出剩余路途中船的位置坐标，只能靠航位推算，根据航向、航速和经过的时间。5月10日，他们登上了南乔治亚岛空无一人的南岸。

沙克尔顿、沃斯利和汤姆·克林横越了岛上山峦重叠的腹地，用了约36个令人精疲力竭的小时，于5月20日抵达斯特罗姆内斯捕鲸站。浮冰逼退/击退了三次救援行动，直到8月30日，“耶乔号”，一艘智利的拖船，才撤出了22个被困于荒岛的人。全部28个坚忍号的成员挺过了他们的这场磨难。

2022年3月5日，搜寻队找到了残骸，位于3,008米深的海底，直立，并且完好无损。它位于沃斯利所记位置以南约四英里处，那个位置记于1915年11月，是小艇出航前几个月测算的；只有一点轻微的差异，对在漂移的冰面上计算出的位置来说，已属难得。

传说理所当然地因沙克尔顿的超凡魅力而称颂他。可这个故事的两头，都取决于沃斯利一丝不苟/十分细心的算数/计算。领导力让28人始终抱成一团。（沃斯利的）精确，不仅将六人送上岸，进入救援范围；并且在一个世纪后，把他们的船带回到世人的视野中。

词汇与表达

departed for v.
驶离了……开往

remote adj.
偏远的

whaling n.
捕鲸

outpost n.
前哨

trapped v.
困住了

vessel n.
船

hull n.
船体
her hull buckling

buckling v.
受压弯曲
her hull buckling

abandoned v.
弃

foundered v.
沉没
foundered on 21 November

logged v.
把……记入了航海日志

stranded adj.
被困住的
The stranded men

endured v.
熬过了

gruelling adj.
艰苦累人的
over five gruelling months

floe n.
浮冰
Their floe split

lifeboats n.
救生艇

bleak adj.
荒凉
a bleak, uninhabited rock

uninhabited adj.
无人居住的

shipping lanes n.
航线

embarked in v.
登上了
and five others embarked in the James Caird, a fragile

embarked on v.
踏上

词汇与表达（续）

fragile adj.
脆弱的

perilous adj.
极度危险的
a perilous voyage

voyage n.
航程

ferocious adj.
狂暴的/猛烈的
As ferocious gales

gales n.
大风

battered v.
多次猛击

snatched v.
抓住了……机会

sights of the sun n.
测日

deducing v.
推断
deducing the rest

dead reckoning n.
航位推算
by dead reckoning

course n.
航向

time elapsed v.
经过的时间

landed on v.
登上了

deserted adj.
空无一人的

traversed v.
横越了
traversed its mountainous interior

mountainous adj.
山峦重叠的

interior n.
腹地

exhausting adj.
令人精疲力竭的

repelled v.
逼退/击退了
Ice repelled three rescue attempts

rescue attempts v.
救援行动

tug n.
拖船

词汇与表达（续）

evacuated v.
撤出了
evacuated the 22 marooned men

marooned adj.
被困于荒岛的
evacuated the 22 marooned men

ordeal n.
磨难
survived their ordeal

located v.
找到了

wreck n.
残骸

upright adj.
直立

intact adj.
完好无损
upright and intact

discrepancy n.
差异
a slight discrepancy

calculation n.
计算

shifting adj.
漂移的

rightly adv.
理所当然地

salutes for v.
因……称颂

charisma n.
超凡魅力
for his charisma

hinge on n.
取决于
hinge on Worsley's

meticulous adj.
一丝不苟/十分细心的
Worsley's meticulous arithmetic

arithmetic n.
算数/计算

Leadership n.
领导力

held together v.
让……抱成一团

Precision n.
精确

ashore adv.
上岸

brought back 表达
把……带回到

文章简述

- 受困：1915年，坚忍号被南极海冰困住，10月弃船、11月沉没；船长沃斯利记下沉船位置，众人自弃船起在浮冰上熬过五个多月。
- 求援：浮冰破裂后，众人乘救生艇抵达远离航线的象岛；沙克尔顿等六人再乘小艇航行约1300公里，前往南乔治亚岛求救。
- 导航：沃斯利航途中仅四次观测太阳定位，其余靠航向、速度和时间推算；登岸后，沙克尔顿、沃斯利等三人又翻山约36小时，到达捕鲸站。
- 获救：浮冰先后逼退三次救援，智利拖船最终于1916年8月接回象岛22人，坚忍号全部28人都活了下来。
- 寻回：2022年，沉船在沃斯利旧记录点以南约四英里处找到；沙克尔顿的领导力让众人团结，沃斯利的精确计算既帮六人求援，也帮后人找到沉船。

一枚邮票改变通信

The Stamp That Changed the Post

一枚邮票怎样简化寄信，又怎样防止重复使用？

英文原文

Before 1840, British **postage** was **levied** according to the number of **sheets** in a letter and the **mileage** of its **carriage**. The sum was usually **payable** on **delivery**, so the person receiving the **correspondence**, not the **sender**, settled it. Prices thus **fluctuated** from one **postal item** to the next.

In 1837 Rowland Hill published a **booklet** urging reform. Parliament **enacted** the **campaigner**'s **proposal** in 1839, and on 10 January 1840 the Uniform Penny Post began. A letter of up to half an **ounce** now cost one **penny** to any **destination** in Britain, and the fee was **prepaid** by whoever sent it. One **flat-rate charge** replaced **variation** with **simplicity**.

The Penny Black, a one-penny **denomination**, was released on 1 May 1840, its **validity** starting on 6 May. **Engraved** on it was a **youthful likeness** of Queen Victoria, the **reigning monarch**. An **aggregate** of 68,808,000 were printed. The sheets had no holes for **separation**, so each stamp was cut out **individually** with **scissors**.

Mailings doubled, from about 80 million **items** in 1839 to over 169 million in 1840. The design had a **drawback**. Against black, the red **cancelling mark** was barely **noticeable**, lacking **contrast**, and the ink could be **erased**, leaving stamps **vulnerable to reuse**. In February 1841 the Penny Red, cancelled in black ink, arrived to **deter tampering** and **fraud**.

As the **initiator** of postage stamps, Britain set a **precedent** and earned an **exemption**. Its stamps are **nameless**, with no country **inscribed** on them. By international agreement, the **sovereign**'s head **suffices for identification**, an **implicit designation** that other countries spell out in print.

中文译文

1840年以前，英国的邮费是根据一封信件中的纸张数量及其运输的里程来征收的。这笔钱通常是在投递时应支付的，所以，收到信件的人，而非寄信人，结清邮费。价格此种方式波动，从一件邮政物品到下一件。

1837年，罗兰·希尔出版了一本小册子，呼吁改革。议会于1839年立法通过了这位改革倡导者的提议，并且在1840年1月10日，“统一便士邮资”开始实行。一封重量不超过半盎司的信，现在寄往英国境内任何目的地都只花费一便士，并且费用由寄信的人预付。一种统一收费以简便取代了差异。

黑便士，一枚面值一便士的邮票，发行于1840年5月1日，它的有效性从5月6日开始。它的上面雕刻着维多利亚女王年轻的肖像，即这位在位的君主。总计有68,808,000枚被印制。这些整版邮票没有用于分离的孔，所以每枚邮票都被单独地用剪刀剪下。

邮寄量翻了一番，从1839年的约8000万件邮寄物品增至1840年的超过1.69亿件。这一设计有个缺点。在黑色底色上，红色注销印记几乎不显眼，由于缺乏反差，且墨水能被擦除，导致邮票容易被重复使用。1841年2月，用黑色墨水盖销的“红便士”问世，旨在防止篡改与欺诈。

作为邮票的首创者，英国开创了先例，并赢得一项豁免。英国邮票是无名的，上面没有刻印国名。根据国际协议，（英国的）君主头像足以用于进行识别，这是一种隐含的标识，然而其他国家则需要把国名印在邮票上。

词汇与表达

postage n.
邮费
British postage was levied

levied v.
征收
levied according to the number

sheets n.
纸张

mileage n.
里程
the mileage of its carriage

carriage n.
运输

payable adj.
应支付的
usually payable on delivery

delivery n.
投递

correspondence n.
信件
receiving the correspondence

sender n.
寄信人

fluctuated v.
波动
Prices thus fluctuated

postal adj.
邮政

item n.
物品

booklet n.
小册子

enacted v.
立法通过了
Parliament enacted

campaigner n.
改革倡导者

proposal n.
提议
the campaigner's proposal

ounce n.
盎司

penny n.
便士

destination n.
目的地

prepaid adj.
预付
the fee was prepaid

flat-rate charge adj.
统一收费
One flat-rate charge

variation n.
差异

simplicity n.
简便

denomination n.
面值
a one-penny denomination

validity n.
有效性
its validity starting on 6 May

Engraved v.
雕刻

youthful adj.
年轻的

likeness n.
肖像

词汇与表达（续）

reigning adj.
在位的

monarch n.
君主

aggregate n.
总计
aggregate of 68,808,000

separation n.
分离

individually adv.
单独地

scissors n.
剪刀

Mailings n.
邮寄量
Mailings doubled

items n.
物品

drawback n.
缺点
The design had a drawback

cancelling mark n.
注销印记
the red cancelling mark

noticeable adj.
显眼

contrast n.
反差
lacking contrast

erased v.
擦除

vulnerable to adj.
容易被
vulnerable to reuse

reuse n.
重复使用

deter v.
防止
to deter tampering and fraud

tampering n.
篡改

fraud n.
欺诈
to deter tampering and fraud

initiator n.
首创者

precedent n.
先例
Britain set a precedent

exemption n.
豁免
earned an exemption

nameless adj.
无名的

inscribed v.
刻印

sovereign n.
君主

suffices for v.
足以用于

identification n.
识别

implicit adj.
隐含的

designation n.
标识

文章简述

- 旧制：英国邮费原按信纸数量和运输距离计算，通常由收信人付款，每封信的价格不一。
- 改革：1840年1月起，希尔推动的改革让半盎司（约14克）以内的信在英国境内统一收一便士，由寄件人预付。
- 邮票：同年5月，印有维多利亚女王头像、面值一便士的黑色邮票发行，称为“黑便士”；当年邮寄量由上年约8000万件增至1.69亿多件。
- 改进：黑票上的红色邮戳不显眼，红墨又容易被洗掉，邮票可能被重复使用；1841年改发红便士并盖黑戳。
- 特例：英国因首创邮票获国际特例，不必印国名，用君主头像就能识别国家；其他国家则要把国名印出。